A Finite Element Julia computational framework involving beams, solid and shell elements for endovascular modelling applications

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Introduction

The use of computational mechanics in biomedical applications presents significant challenges, particularly in scenarios where biological tissues (vascular wall, thrombi or plaque) interact with artificial, wire-like medical devices. Accurately modeling the contact between these entities is complex due to their vastly different mechanical properties and structural conception. Recent advancements have introduced a finite element method (FEM) computational model for simulating the deployment of flow diverters [1, 2]. The model utilizes corotational beam theory to simulate the metallic wires of flow diverters, which becomes highly efficient. As a continuation of the above work, we present the development of a computational framework for contact between endovascular devices (modelled using corotational beams), vascular wall (modelled using rotation-free shell elements, see [3]) and thrombi/plaque (modelled using hyperelastic tetrahedron elements). The implementation is carried out in the Julia programming language

Methods

A fully implicit method involving beam elements, shells and solids is presented. The beam elements are modelled using a previously reported corotational beam formulation [1,2]. Shell elements are modelled using a rotation-free nonlinear shell that relies on the use of an element patch which allows for capturing the curvature tensors required for bending stiffness [3]. Finally, the solid element is modelled using linear compressible tetrahedrons. Contact between elements is defined by performing master-slave strategy and using a penalty energy function, using both line to surface and point to triangle contact. The system of equations is integrated in time using a fully implicit HHT-alpha method, with the respective residual contributions adequately linearized and added as contributions to the Newton-Raphson tangent matrix [1,2]. An update strategy is carried out to find, at periodic time steps, a limited number of contact pairs in order to limit the non-zero terms of the sparsity matrix.

Results

Figure 1 shows the simulation results of a stent, modelled using corotational beams, and an arterial segment, modelled with shell elements as well as the results of modelling a stent against a plaque using linear compressible tetrahedrons. Figure 2 shows the conservative properties of the time integration algorithm.

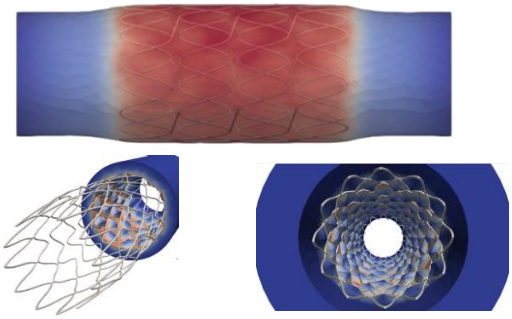


Figure 1: Top: stent against an artery modelled using shell elements. Bottom: stent against plaque using compressible tetrahedrons.

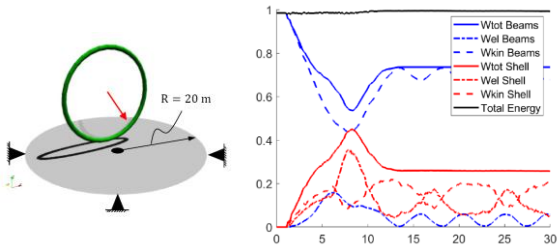


Figure 2: Ring against a flexible floor. Problem setup (left) and conservation of energy of the beam, shell and system (right).

Conclusions

The computational framework shows a promising comptutational alternative for modelling endovascular devices.

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